



Blockchain transaction fee and Ethereum Merge

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ABSTRACT

We study the determinants of transaction fees in the Bitcoin and Ethereum blockchains, particularly focusing on the recent Merge – Ethereum's transition from a Proof-of-Work to a Proof-of-Stake consensus mechanism. Using multivariate regression analysis, we find that blockchain fees in USD increase with network congestion, measured by the mempool count, in a non-linear manner as hypothesized by Huberman et al. (2021). Additionally, the fees are positively influenced by the respective native currency's returns over the past six months. The Merge resulted in a fee reduction for both Bitcoin and Ethereum blockchains.

1. Introduction

Cryptocurrencies are decentralized payment systems based on a consensus mechanism to validate transactions, such as Proof-of-Work (PoW) or Proof-of-Stake (PoS). In PoW, *miners* must solve computational puzzles to include a block of transactions in the blockchain. Those with greater computational power stand a better chance of validating blocks and earning a reward, making this mechanism energy intensive. Conversely, in PoS, *validators* stake the cryptocurrency's native token. Those with the highest stakes are most likely to be selected to produce the next block. This process significantly reduces the network's power demand. Recently, on September 15, 2022, Ethereum transitioned from PoW to PoS in a highly anticipated Merge, while Bitcoin retains its PoW consensus mechanism.

Unlike centralized systems with relatively stable transaction fees, blockchain transactions fees can fluctuate significantly. Typically, mining reward consists of both a fixed and a variable component, with the latter determined by the transaction fee chosen by network users. Transaction fees are affected by network congestion and upgrades. Stability in transaction fees is one of the key factors that would allow further integration of cryptocurrencies into the economy. Fees impact transaction originator and validator participation alike. Therefore, our objective is to analyze the factors determining transaction fees and the effects of the Merge on Ethereum and Bitcoin transaction fees.

In contrast to traditional stock markets, where dealers can increase liquidity supply in response to increased demand, limitations on block size and block times can cause network congestion. This arises when the rate of transaction submission outpaces processing capacity.¹ When numerous transactions flood the network, users tend to offer higher fees to prioritize their transactions, thereby providing miners with incentives to prioritize transactions with the highest fees. We use the mempool count as a measure for

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¹ A Bitcoin block has a size limit of 1 MB. Each Ethereum block has a target size of 15 million gas units (and a limit of 30 million gas units).

congestion, which represents the number of pending transactions waiting to be added to the blockchain. The more transactions are in a queue, the longer the expected time for fulfillment. We find that the fees for both the Bitcoin and Ethereum blockchain are positively related to mempool count. For Bitcoin, we find that the positive impact of mempool count on fees is significant only after the mempool count reaches 80th percentile and is strongest when mempool count is in the highest decile. For Ethereum, the effect begins around the 40th percentile, intensifying as the mempool count increases. Blockchain fees for both Ethereum and Bitcoin are positively related to past six-month returns.

This study is related to the emerging literature examining transaction fees for cryptocurrencies. To our knowledge, three studies have developed theoretical models that positively link congestion to transaction fees, in particular when the network is approaching capacity. Huberman et al. (2021) present a theoretical model of the Bitcoin payment system, showing that its construction protects users from monopoly pricing due to competition and free entry and that transaction fees increase with congestion in a non-linear manner. Easley et al. (2019) link Bitcoin transaction fees positively to mean waiting time through a game-theoretic model. Chiu and Koepl (2019) theoretically examine the relationship between Bitcoin transaction fees, block size, and user characteristics. Their simulation results suggest that users are willing to pay more when their transactions are more urgent and when the block size is smaller. Consistent with these theoretical models, we show that transaction fees grow nonlinearly with mempool count, although the dynamics are slightly different for Ethereum and Bitcoin.

Recent empirical research has shed light on fee determinants in cryptocurrency networks. Brown et al. (2022) examine Bitcoin, exploring congestion effects and the adoption of the Segwit protocol, with a focus on fee sustainability. Meanwhile, Donmez and Karaivanov (2022) empirically study Ethereum's transaction fees, finding a positive and non-linear relationship with block utilization and transaction type, though their study predates the EIP-1559 upgrade in August 2021. This upgrade transformed Ethereum's fee structure by introducing a base fee that is burned to enhance fee predictability and variable-sized blocks (Roughgarden, 2020).

Due to constant innovations in blockchains, the interactions and dynamics within these systems undergo significant changes, altering relationships and incentives. Consequently, to ensure that our understanding remains relevant and accurate, we focus our analysis on the newest upgrade – the Ethereum Merge. While the Merge's primary aim was to enhance Ethereum's energy efficiency, provide future scalability, and lower barriers for network participants, it was not predicted to directly influence transaction fees. However, the reduced cost of the consensus mechanism, eliminating the need for expensive computing hardware, suggests validators might now accept lower fees.

In a recent study, Kapengut and Mizrach (2023) show that the Ethereum's transition to PoS significantly reduced energy consumption by 99.98 percent and they note an increase in the fee denominated in Ether in a univariate setting over a two-month window. We substantially extend their analysis in several ways. First, we perform our analysis in a multivariate setting using robust specifications and controlling for the most important determinant of fees - congestion. Second, we extend the analysis window to a three year period surrounding the event to avoid noise around the Merge, especially when there was doubt if the Merge would be successful.² Third, we focus on economically important fees in U.S. Dollars (USD), consistent with investors comparing fees across assets as well as considering exchange and wallet fees expressed in one currency.

We find that transaction fees in USD decreased for both Ethereum and Bitcoin blockchains, with a much more substantial decrease for the former. Although the Merge does not directly impact the Bitcoin blockchain, the two blockchains' transaction fees tend to move together. A shift in one might precipitate a change in the other.

Our study contributes to recent Merge-related research. De Vries (2023) highlights a 99.99 percent reduction in power consumption and discusses Merge's applicability to other cryptocurrencies. Cortes-Goicoechea et al. (2023) propose strategies to optimize validator rewards and network efficiency, addressing factors like transaction order and gas fees. Grandjean et al. (2023) reveal high validator participation but express concern about power concentration that can endanger network security. Our focus extends to Merge's impact on transaction fees in both Ethereum and Bitcoin blockchains.

While existing literature on the costs of financial intermediation in traditional finance is vast, the literature on transaction costs in decentralized finance is still sparse. To the best of our knowledge, there has been no empirical study combining the factors that impact blockchain fees and the Ethereum Merge and our study aims to fill this gap.

The rest of the paper proceeds as follows. Section 2 provides data and descriptive statistics. Section 3 outlines our estimation methodology and discusses the impact of mempool count and the Merge on transaction fees. Section 4 concludes.

2. Data and descriptive statistics

We download daily closing prices in USD, daily average transaction fees in USD and mempool count for Ethereum from <https://etherscan.io/>, and the same variables for Bitcoin from <https://blockchain.com/>.

Ethereum fee measures the daily average fee paid in USD when an Ethereum transaction is processed by a miner or validator and confirmed. It consists of the base fee set by the protocol and a priority tip earned by validators. The base fee gets removed from circulation when a new block is created. The priority tip is set by the user depending on how fast they need the transaction to be completed. Bitcoin fee measures the daily average fee in USD paid to miners when the transaction is included in the blockchain – it is also set by the user based on the urgency of adding their transaction to the blockchain. This does not include a fixed block reward earned by the miner, which comes from newly minted Bitcoin. Although transaction fees can vary intraday, this study only focuses on

² The number of protocols that would migrate to the new consensus blockchain was not known precisely until it became clear that the Merge was a successful event. <https://www.coindesk.com/markets/2022/08/15/ether-derivative-token-suggests-the-merge-is-not-fully-priced-in-yet/>

daily averages.

We choose to analyze transaction fees expressed in USD as investors generally compare transaction costs across various cryptocurrencies and other assets or across other means of money transfers in dollars. Additionally, investors frequently consider exchange transaction fees and wallet fees, also denominated in USD. Further, miners analyze fees in USD as they rely on purchasing mining resources in USD (Huberman et al., 2021).

The mempool is defined as the average number of pending transactions per day. For a transaction to be confirmed, it needs to be included in a block. The speed of inclusion depends on the fee offered by the user. The mempool is considered a good proxy for congestion,³ as a high number of pending transactions leads to increased traffic, resulting in longer confirmation time and higher fees.⁴

Our multivariate analysis spans a period of three years from April 14, 2020, to April 14, 2023. The univariate analysis around the Merge on September 15, 2022, focuses on a year surrounding the event from March 15, 2022, to March 15, 2023.

In Table 1 Panel A, we present the mean, median, and the standard deviation of transaction fees, mempool count, six-month returns, and the price for Bitcoin and Ethereum for our sample period. The average transaction fee for Bitcoin (Ethereum) during our sample period is USD 4.88 (11.36). The mean values are higher than the median, suggesting positive skewness in the distribution. We observe high volatility, as measured by standard deviation, in the fees of both Bitcoin and Ethereum, with both showing similar variability relative to their mean. Average fees for cryptocurrency transactions vary due to network congestion, demand fluctuations driven by regulatory announcements, exchange outages, network upgrades, and speculative activity. The average six-month returns are positive in our sample period.

In Panel B, we present the correlation matrix for these variables. Bitcoin (Ethereum) transaction fee is positively correlated to Bitcoin (Ethereum) mempool count in line with theoretical predictions that higher network congestion driven by increased demand relates to higher fees. The fees are also positively correlated with native currency's six-month return, which can be explained by increase in the number of transactions following strong past returns.

We also note a positive correlation between Ethereum and Bitcoin fees which can be linked to the price and demand sensitivity of both cryptocurrencies to similar external macroeconomic and regulatory factors, market sentiment, and technical events. This motivates us to test if the Merge affected Bitcoin fees as well.

In Table 2 we present these numbers separately for a six-month period before and after the Ethereum Merge (September 15, 2022). We find that the average transaction fee for Bitcoin (Ethereum) blockchain went down from USD 1.58 (8.50) to USD 1.24 (3.50). The percentage decrease in the fees is significantly higher (58.8 percent) for Ethereum blockchain as compared to Bitcoin blockchain's decrease (21.5 percent). Mempool count decreased only for Ethereum, which, pointing to lower congestion, could also have contributed to the decreasing fees. Six-month returns went up for both cryptocurrencies (despite a strong price decline in late 2022).

We plot the mempool count, the transaction fee and a five-day moving average of the price in Fig. 1a for Bitcoin and in Fig. 1b for Ethereum. The co-movement of transaction fees with mempool count is very clear for Bitcoin, and less so for Ethereum. In Fig. 1a we also notice a clear increase in the Bitcoin mempool count (and fees) after January 21, 2023, when the new Ordinals Protocol allowing for the creation of non-fungible assets was launched. It appears that Ethereum fees are more driven by the currency price in USD as compared to Bitcoin, as also shown by the correlation matrix.

Taken together, the univariate results presented above point to the importance of both mempool count and currency returns for the transaction fees. It is imperative to control for these factors as we analyze the impact of the Merge on the fees of Bitcoin and Ethereum. In the next section, we use a multivariate regression model to formally test our preliminary findings from the univariate results.

3. Estimation methodology and results

We now turn to studying determinants of transaction fees, including the effects of the recent Merge. As postulated theoretically by Chiu and Koepl (2019); Easley et al., (2019) and Huberman et al., (2021), transaction fees should be positively related to network congestion and more strongly so when network is at capacity, implying a nonlinear relationship. The capacity for processing transactions on both Bitcoin and Ethereum networks is constrained due to block size and time limitations, resulting in congestion. As mempool gets larger, only those transactions with the highest fees are validated by the miners/validators (Donmez and Karaivanov, 2022). At the same time, users with high delay costs are willing to offer higher fees to get their transaction prioritized (Huberman et al., 2021). Moreover, for transaction fees expressed in USD, times of high congestion indicate a positive shock to the demand, driving the cryptocurrency price up and consequently the fee in USD increases as well. Finally, in an environment of higher volatility, users are motivated to complete transactions faster and are willing to pay a higher fee for that (Brown et al., 2022).

We also hypothesize that the Merge, transitioning Ethereum from the PoW to PoS mechanism resulted in a decrease in fees in Ethereum. The change in consensus mechanism was not designed to affect the fees. Its primary objective was energy efficiency and preparing the network for future upgrades, including scalability – a potential solution to capacity limitations. Due to the change in consensus mechanism, miners who used to mine for the Ethereum blockchain using energy consuming Graphics Processing Units (GPUs) were replaced by the validators who stake the tokens they own. Since the cost of consensus mechanism decreased substantially, as no expensive computing equipment is required, it is possible that validators would be willing to accept a lower fee. We define the Merge event dummy as having a value of 1 after September 15, 2022, and 0 otherwise.

Based on our univariate results, we control for past cumulative six-month return, as growing cryptocurrency returns attract more

³ Mempool is discussed or used as a proxy for congestion by Easley et al. (2019), Huberman et al. (2021) and Brown et al. (2022).

⁴ <https://www.blockchain.com/explorer/charts/mempool-count>

Table 1

Full sample descriptive statistics and correlation.

Panel A: Descriptive statistics							
<i>N</i> = 1096	<i>Bitcoin</i>			<i>Ethereum</i>			
	Mean	Median	Std Dev	Mean	Median	Std Dev	
Transaction fees	4.88	2.28	7.07	11.36	5.15	14.11	
Mempool count	15,871.93	6,517.14	20,227.80	156,699.83	160,201.00	28,552.31	
Six-month return	47.56 %	11.34 %	118.69 %	91.20 %	52.95 %	159.31 %	
Price in USD	30,378.93	27,936.83	15,916.04	1,827.20	1,651.07	1,176.69	
Panel B: Correlation matrix							
Variable	Bitcoin transaction fee	Bitcoin mempool count	Bitcoin six-month return	Bitcoin price	Ethereum transaction fee	Ethereum mempool count	Ethereum six-month return
Bitcoin mempool count	0.7803***						
Bitcoin six-month return	0.8127***	0.6372***					
Bitcoin price	0.4225***	0.1062***	0.4369***				
Ethereum fee	0.1474***	−0.0415	0.1486***	0.7231***			
Ethereum mempool count	0.0356	−0.0597**	−0.0320	0.3222***	0.1717***		
Ethereum six-month return	0.7808***	0.5481***	0.8751***	0.4317***	0.1797***	−0.0712**	
Ethereum price	0.0556*	−0.2133***	−0.0041	0.8738***	0.7471***	0.3482***	0.1032***

We provide summary statistics of blockchain transaction fees, prices, and returns of native currency in USD, as well as mempool count. **Panel A** provides mean, median, and standard deviation of these variables for both Bitcoin and Ethereum blockchain. **Panel B** provides correlation matrix for these variables. We have omitted correlation of a variable with itself from the table. Three stars denote significance at 1% or less, two stars denote significance at 5% or less; one star denotes significance at 10% or less.

Table 2

Univariate analysis – descriptive statistics before and after the Merge.

	<i>Bitcoin</i>			
	Before (<i>N</i> = 183)	After (<i>N</i> = 183)	Difference	T-stat
Transaction fees	1.58	1.24	−0.34***	−7.67
Mempool count	5,414.1	6,370.3	956.1	1.46
Six-month return	−38.6 %	−27.4 %	11.2 %***	5.14
Price in USD	29,885.3	19,626.6	−10,258.8***	−14.9
	<i>Ethereum</i>			
	Before (<i>N</i> = 183)	After (<i>N</i> = 183)	Difference	T-stat
Transaction fees	8.50	3.50	−5.00***	−4.37
Mempool count	182,709.0	163,572.8	−19,136.3***	−8.99
Six-month return	−41.2 %	−20.6 %	20.6 %***	8.19
Price in USD	2,018.0	1,405.7	−612.3***	−11.2

We compare averages of blockchain transaction fees, returns, prices in USD and mempool count for a 6-month period before and a 6-month period after the Merge on September 15, 2022, when Ethereum transitioned from PoW to PoS. Three stars denote significance at 1% or less.

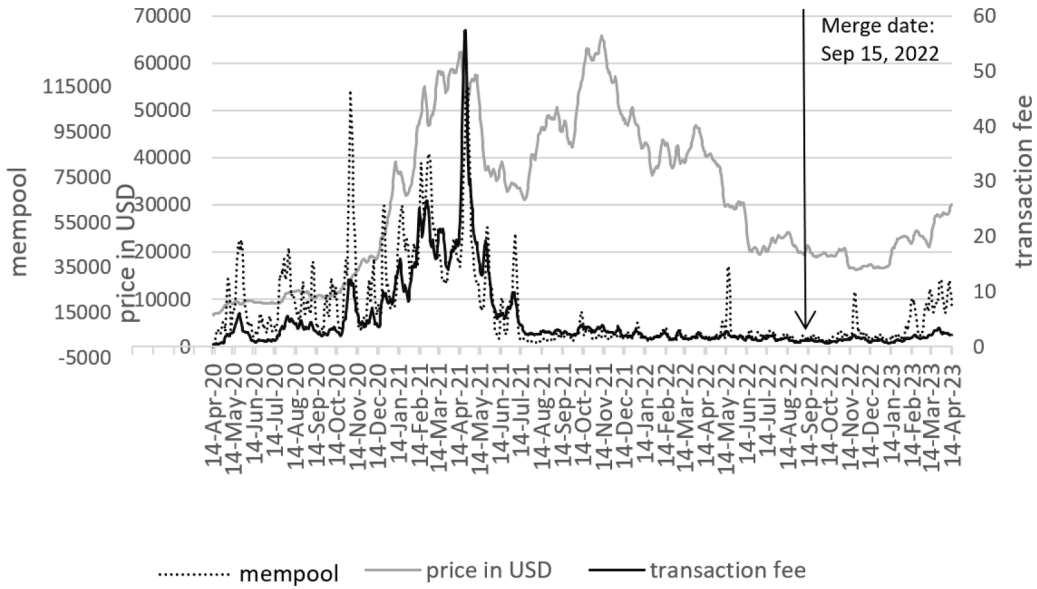


Fig. 1a. Bitcoin.

In this figure, we plot a five-day moving average of the bitcoin price, mempool count, and transaction fee in USD from April 14, 2020 to April 14, 2023.

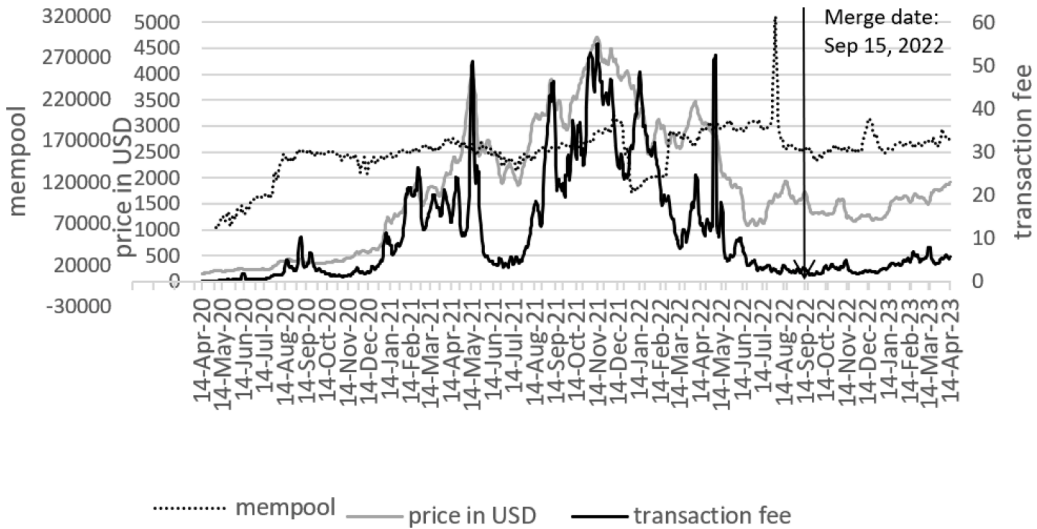


Fig. 1b. Ethereum.

In this figure, we plot a five-day moving average of the Ethereum price, mempool count, and transaction fee in USD from April 14, 2020 to April 14, 2023.

users, spur speculative behavior and in turn can drive up the fees as well.

We aim to test the following equation for Bitcoin and Ethereum separately:

$$\text{Transaction fee}_t = \alpha + \beta_1 \text{Event}_t + \beta_2 \text{Six month return}_t + \beta_3 \text{Mempool}_t + \epsilon_t \quad (1)$$

Given that this time-series specification runs the risk of heteroskedasticity and autocorrelation, we perform a series of tests to ensure appropriate estimation approach.

First, we perform unit root tests on the transaction fee variables and confirm that both are stationary.⁵ Then we test the OLS

⁵ Augmented Dickey-Fuller test for transaction fee renders Mackinnon p-value of Z(t) -11.078*** for Ethereum and Z(f)=-3.974*** for Bitcoin (***) denotes significance at 1 percent level).

specification from Eq. (1) and confirm heteroskedasticity and autocorrelation in residuals.⁶ To correct for these issues, we use heteroskedasticity and autocorrelation consistent (HAC) standard errors. In Table 3, columns 1 and 3 we present the results of estimating the Eq. (1) using OLS with HAC estimator of standard errors based on quadratic spectral kernel (Andrews, 1991). We also verify that the results hold with the HAC estimator based on Newey-West method.

In columns 1 and 3 examining Bitcoin fees and Ethereum fees respectively, we find that the coefficient of mempool is positive and significant. Since we expect a nonlinear relationship between the transaction fee and the mempool, we replace the mempool count variable with deciles of our sample-days based on the mempool count and create a dummy variable for each decile rank. We then re-estimate the equation with HAC standard errors in columns 2 and 4. In the Bitcoin regression in column 2, only deciles 9 and 10 are significantly positive. The fee is higher by USD 2.13 and USD 8.35 for decile 9 and decile 10, respectively, as compared to the lowest decile. It is clear than when the Bitcoin mempool approaches capacity, its count starts affecting the fees. In column 4 presenting Ethereum regression, we find that the dummies for deciles 6 through 10 are all positive and significant. Ethereum transaction fees for these deciles increase monotonously until decile 9 and decline for decile 10. The coefficients for deciles 6 through 10 are USD 8.27, USD 12.22, USD 13.40, USD 16.92, and USD 10.57, respectively. This result suggests that the blockchain fee is increasingly sensitive to mempool count for Ethereum. Our findings are consistent with theoretical priors (Chiu and Koepl, 2019; Easley et al., 2019 and Huberman et al., 2021) that larger mempool count results in a higher transaction fee with the relationship significantly stronger when mempool is large. Increase in fees driven by the mempool count is consistent with blockchain users willing to pay higher transaction fees for immediacy, to reduce delay cost, and to ensure fulfillment in a volatile environment.

This finding is crucial for users aiming to forecast transaction costs and associated risks. With a high mempool count, users run the risk of their transactions remaining unconfirmed for extended periods, leading to potential lost profit opportunities. Conversely, less experienced users may be prone to overpaying in fees. Finally, high and volatile fees during times of high congestion can pose a barrier to entry for new users.

We now turn to the impact of the Merge on the transaction fees. In columns 1 and 2 the coefficient on the event dummy is negative and significant, implying Bitcoin blockchain fee decreasing by USD 1.02 and USD 0.95 respectively. This supports the notion that the Merge indirectly affected Bitcoin, given the positive correlation between Ethereum and Bitcoin fees and their comparable sensitivity to external factors.

The coefficient of the event dummy is negative and significant in both columns 3 and 4. This implies that post-Merge, the Ethereum blockchain fee is lower by USD 9.02 (USD 9.82) in column 3 (4), and this fee change is about nine times higher than the fee change for Bitcoin. The decrease in Ethereum fees, while surprising, is not unfounded. The new conditions make it more conducive for validators to accept lower rewards. After the Merge, the cost of validation decreased significantly because specialized Graphical Processing Units (GPUs) requiring high energy are no longer needed. Validator rewards are now directly proportional to the amount of Ethereum staked and the block times are set at fixed 12 s. In contrast, under the PoW mechanism, the average block time was 13 s with some variance. This leads to more predictable validation rewards. Upcoming Ethereum network upgrades will focus on scalability improvements through layer 2 rollups and Danksharding,⁷ which are expected to systematically reduce fees.

Our findings of a decrease in fees following the Merge contrast with the results presented by Kapengut and Mizrach (2023). They show fees denominated in Ether increasing using a 2-month window surrounding the Merge. We believe our approach provides a more comprehensive and robust analysis using multivariate regression with HAC errors and controlling for congestion and past returns. We also extend the window of analysis to three years to eliminate impact of noise and uncertainty around the Merge. Finally, we focus on economically relevant fees in USD.

As postulated, fees for both Bitcoin and Ethereum are positively related to past 6-month returns. This aligns with the positive correlation between them presented in Table 1, Panel B. It also supports the observation that fees, when denominated in USD, increase as the cryptocurrency's value appreciates. Positive returns typically attract more attention from investors and can amplify bullish sentiment leading to increased demand and rise in speculative trading. This results in more transactions, causing network congestion and consequently, elevated fees. Furthermore, as users earn higher returns, they may become less sensitive to transaction fees.

While Bitcoin blockchain only has one transaction type (transfer of Bitcoin from one public address to another public address), Ethereum blockchain allows for more complex transactions such as smart contract deployment transaction or execution of a smart contract. For this reason, we find that the explanatory power of the Bitcoin regressions, as measured by the adjusted R squared, is quite high (66 – 73 percent), while it is much lower for Ethereum (13 – 25 percent).

To ensure robustness of our results, we consider a potential endogeneity problem with regards to the relationship between

⁶ Summary for heteroskedasticity and autocorrelation test statistics.

Test	specification for Ethereum	specification for Bitcoin	outcome
Breusch-Pagan	Chi2 109.95***	Chi2 of 2431.88***	heteroskedasticity in residuals for both
White's	Chi2 16.53***	Chi2 of 429.95***	
Durbin Watson	d-statistic (4,1096) = 0.47	d-statistic (4, 1096) = 0.22	residuals autocorrelated in both
Breusch-Godfrey	Chi2 of 639.11***	Chi2 of 871.28***	

⁷ Initially the scalability was intended to be increased via sharding - partitioning each block into smaller subsets. However, the updated Ethereum upgrade roadmap focuses on layer 2 optimistic rollup protocols intended to increase throughput of the base layer. This will be further enabled by Danksharding, a novel concept of sharding.

Table 3
Multivariate analysis – determinants of blockchain fees and the effect of the merge.

Variable	Bitcoin fee		Ethereum fee	
	(1)	(2)	(3)	(4)
Intercept	1.1618*** [3.22]	2.5321*** [84.23]	-4.6998 [-0.53]	4.3898 [1.46]
Event	-1.0222** [-2.01]	-0.9483*** [-3.00]	-9.0185*** [-8.92]	-9.8242*** [-14.22]
Six-month return	3.0021*** [11.58]	3.7644*** [18.33]	1.6292*** [5.57]	1.9213*** [4.44]
Mempool	0.0002** [13.12]		0.0001** [2.11]	
Decile 10		8.3513*** [8.22]		10.5715*** [2.58]
Decile 9		2.1277*** [4.06]		16.9194*** [3.61]
Decile 8		0.8058* [1.91]		13.4019*** [3.01]
Decile 7		-0.3690 [-1.10]		12.2190*** [2.93]
Decile 6		-0.0590 [-0.14]		8.2685** [2.39]
Adjusted R Squared	0.728	0.664	0.132	0.251
Wald chi2(3)	227,298***	1270,000,000***	261,371***	15,400,000***
Number of Observations	1,096	1,096	1,096	1,096

We run a multivariate regression following Eq. (1) with Bitcoin and Ethereum fee as the independent variables in columns 1 and 3 and mempool count, six-month return, and Merge event dummy as dependent variables. In columns 2 and 4, we replace the mempool count by the decile dummies for mempool count. We only report the result for the top 5 decile dummies. T-statistics are based on heteroskedasticity and autocorrelation consistent standard errors using quadratic spectral kernel (Andrews, 1991). Three stars denote significance at 1% or less, two stars denote significance at 5% or less; one star denotes significance at 10% or less.

mempool count and transaction fee. While a larger mempool increases fees due to users offering higher fees to receive priority, an increase in fees can, in turn, affect demand and mempool size. In unreported results, following Brown et al., (2022), we use an instrumental variable approach with HAC standard errors, where one day lagged mempool serves as an instrument to concurrent mempool count. The results of the IV regression remain consistent. Our results also hold when we use a 3-month lagged return instead of a 6-month lagged return.

4. Conclusion

This study provides an analysis of the determinants of blockchain transaction fees, placing a particular emphasis on the effects of Ethereum's transition from Proof-of-Work to Proof-of-Stake. Ethereum and Bitcoin USD fees increase with mempool count, especially once the mempool count is large. The fees also depend positively on the native currency's past returns. Recent Ethereum Merge resulted in decreased USD fees for Ethereum as well as Bitcoin.

Historically, blockchain transaction fees have been volatile and at times extremely high, especially for more complex interactions such as smart contracts in Ethereum. Understanding their determinants can be helpful for investors and users assessing the risk and impact of fees on their potential return, as well as for risk managers looking for opportunities to hedge against fee volatility. Regulatory environment is also likely to change. Possibly, more currencies will move to PoS. All these changes may provide a rich setting for future research. An extension of this study can investigate determinants of transaction fee based on the type of transaction – smart contracts vs. token transfer. An intraday analysis could provide further insights as well.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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